

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 1/10 - U_MCU

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

U_PD - CH224K

15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1

Creates 3.3 V from diode-OR USB/PD power.
Low-current LDO retained for prototype only;
measure input spikes, current and temperature.

U_DRV - A4988

Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6

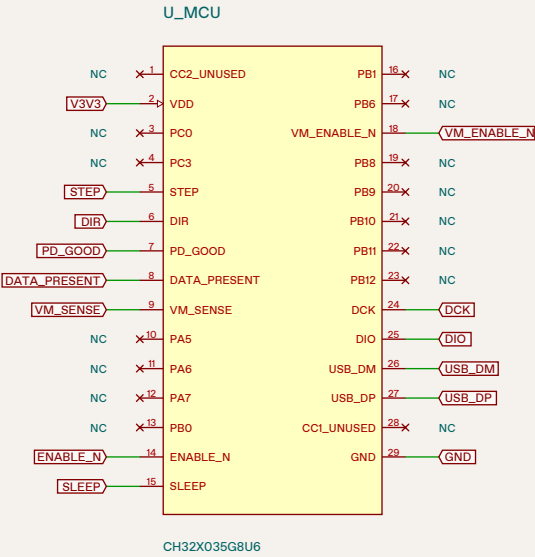
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125

Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F

Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.



Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 2/10 - U_DRV

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

U_PD - CH224K

15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1

Creates 3.3 V from diode-OR USB/PD power.
Low-current LDO retained for prototype only;
measure input spikes, current and temperature.

U_DRV - A4988

Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

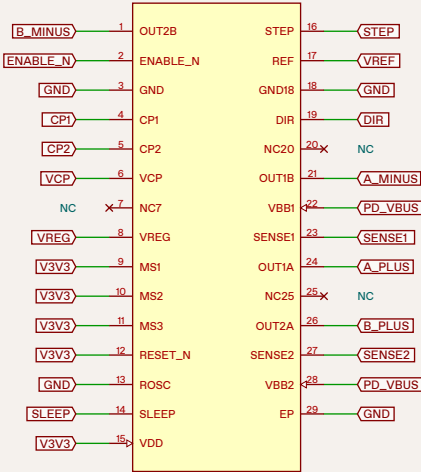
U_VM_ISO - SN74CBTLV1G125

Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F

Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

U_DRV



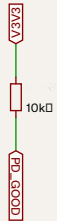
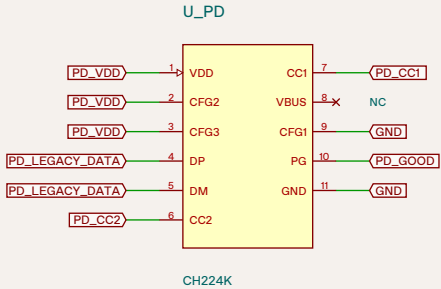
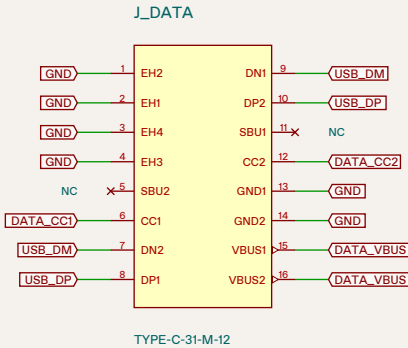
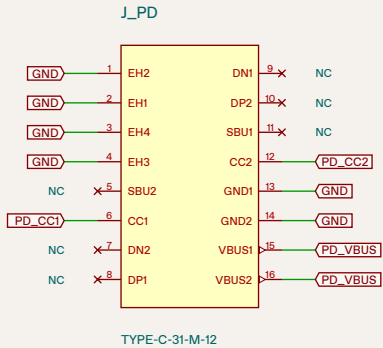
A4988SETTR-T

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 3/10 - J_PD, J_DATA, U_PD, R_PD, C_PD, R_PG



COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR. Checks PD voltage before enabling the motor. Internal clock avoids an external crystal.

U_PD - CH224K

15 V request; CFG1=0, CFG2=CFG3=1. PG is open-drain, active low. PD-only mode: chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1

Creates 3.3 V from diode-OR USB/PD power. Low-current LDO retained for prototype only; measure input spikes, current and temperature.

U_DRV - A4988

Fixed 1/16 steps; 10k enable/sleep defaults. 3.9k/1k REF divider, 0.24 ohm sense resistors: 0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6

Clamps ESD on USB D+ and D- at the data port. VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125

Isolates VM ADC when logic power is off. 10k OE pull-up defaults off; MCU gates ON. Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F

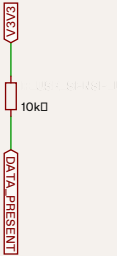
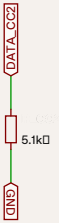
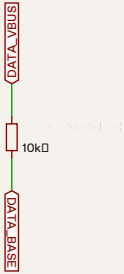
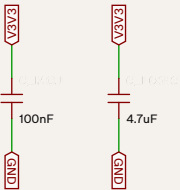
Detects host 5 V without direct GPIO feed. 10k base feed and 10k 3.3 V collector pull. DATA_PRESENT LOW means host attached.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 4/10 - C_MCU, C_LOGIC, R_CC1, R_CC2, R_USB_SENSE_H, R_USB_SENSE_L



COMPONENT PURPOSE

U_MCU - CH32X035G8U6
USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

U_PD - CH224K
15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1
Creates 3.3 V from diode-OR USB/PD power.
Low-current LDO retained for prototype only;
measure input spikes, current and temperature.

U_DRV - A4988
Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125
Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

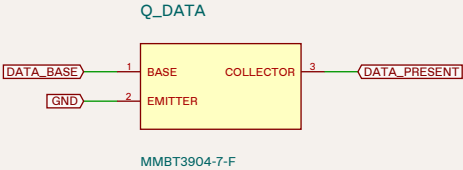
Q_DATA - MMBT3904-7-F
Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 5/10 - Q_DATA, R_VM_H, R_VM_L, U_VM_ISO, R_VM_EN, R_VM_BLEED



COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR. Checks PD voltage before enabling the motor. Internal clock avoids an external crystal.

U_PD - CH224K

15 V request: CFG1=0, CFG2=CFG3=1. PG is open-drain, active low. PD-only mode: chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1

Creates 3.3 V from diode-OR USB/PD power. Low-current LDO retained for prototype only; measure input spikes, current and temperature.

U_DRV - A4988

Fixed 1/16 steps; 10k enable/sleep defaults. 3.9k/1k REF divider, 0.24 ohm sense resistors: 0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6

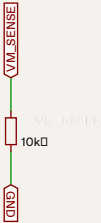
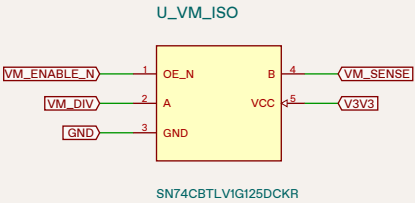
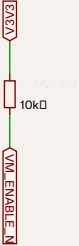
Clamps ESD on USB D+ and D- at the data port. VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125

Isolates VM ADC when logic power is off. 10k OE pull-up defaults off; MCU gates ON. Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F

Detects host 5 V without direct GPIO feed. 10k base feed and 10k 3.3 V collector pull. DATA_PRESENT LOW means host attached.

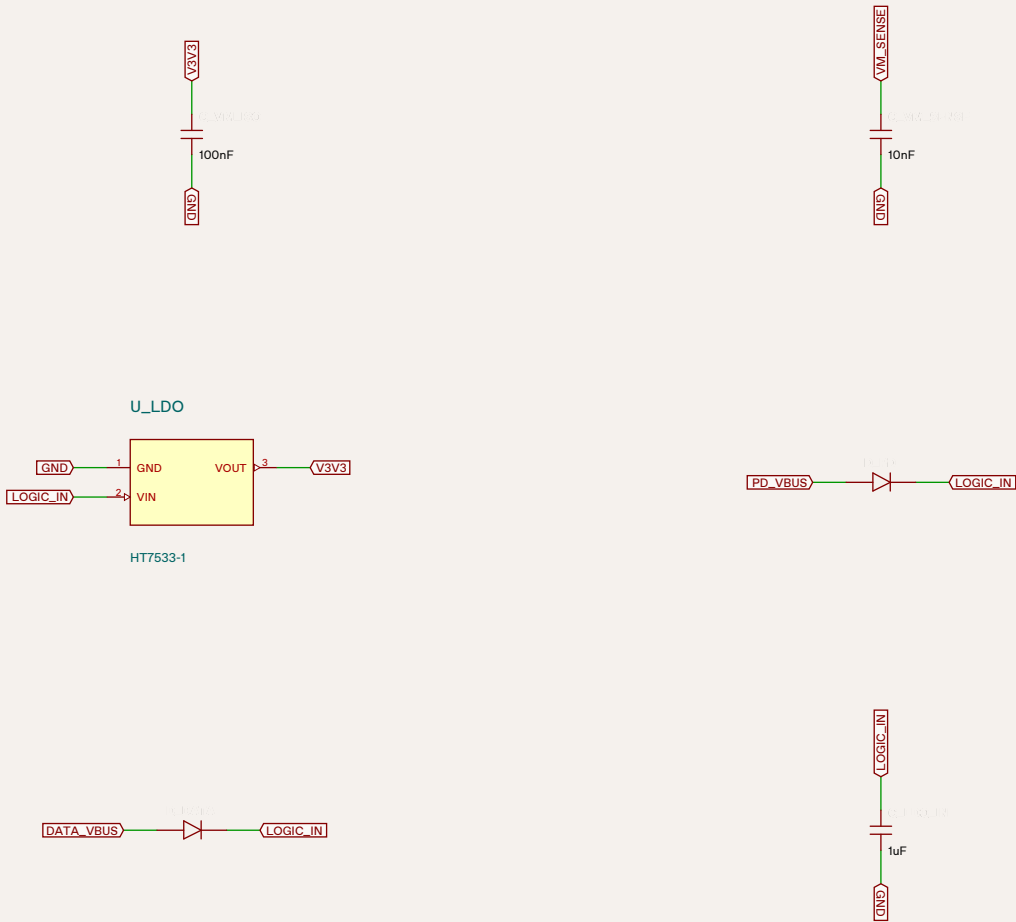


Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 6/10 - C_VM_ISO, C_VM_SENSE, U_LDO, D_PD, D_DATA, C_LDO_IN



COMPONENT PURPOSE

U_MCU - CH32X035G8U6
USB computer control; generates STEP/DIR. Checks PD voltage before enabling the motor. Internal clock avoids an external crystal.

U_PD - CH224K
15 V request: CFG1=0, CFG2=CFG3=1. PG is open-drain, active low. PD-only mode: chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1
Creates 3.3 V from diode-OR USB/PD power. Low-current LDO retained for prototype only; measure input spikes, current and temperature.

U_DRV - A4988
Fixed 1/16 steps; 10k enable/sleep defaults. 3.9k/1k REF divider, 0.24 ohm sense resistors: 0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6
Clamps ESD on USB D+ and D- at the data port. VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125
Isolates VM ADC when logic power is off. 10k OE pull-up defaults off; MCU gates ON. Bus/Z1 when ON; wait 0.5 ms then sample.

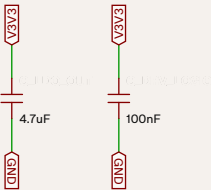
Q_DATA - MMBT3904-7-F
Detects host 5 V without direct GPIO feed. 10k base feed and 10k 3.3 V collector pull. DATA_PRESENT LOW means host attached.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 7/10 - C_LDO_OUT, C_CP, C_VCP, C_VREG, C_DRV_LOGIC, C_VM



COMPONENT PURPOSE

U_MCU - CH32X035G8U6
USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

U_PD - CH224K
15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1
Creates 3.3 V from diode-OR USB/PD power.
Low-current LDO retained for prototype only;
measure input spikes, current and temperature.

U_DRV - A4988
Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125
Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

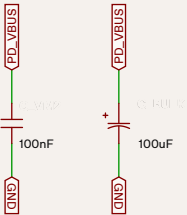
Q_DATA - MMBT3904-7-F
Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 8/10 - C_VM2, C_BULK, R_SA, R_SB, R_REF_H, R_REF_L



COMPONENT PURPOSE

U_MCU - CH32X035G8U6
USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

U_PD - CH224K
15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1
Creates 3.3 V from diode-OR USB/PD power.
Low-current LDO retained for prototype only;
measure input spikes, current and temperature.

U_DRV - A4988
Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125
Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F
Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 9/10 - C_REF, R_ENABLE, R_SLEEP, J_MOTOR, J_DEBUG, U_ESD

COMPONENT PURPOSE

U_MCU - CH32X035G8U6

USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

U_PD - CH224K

15 V request; CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1

Creates 3.3 V from diode-OR USB/PD power.
Low-current LDO retained for prototype only;
measure input spikes, current and temperature.

U_DRV - A4988

Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6

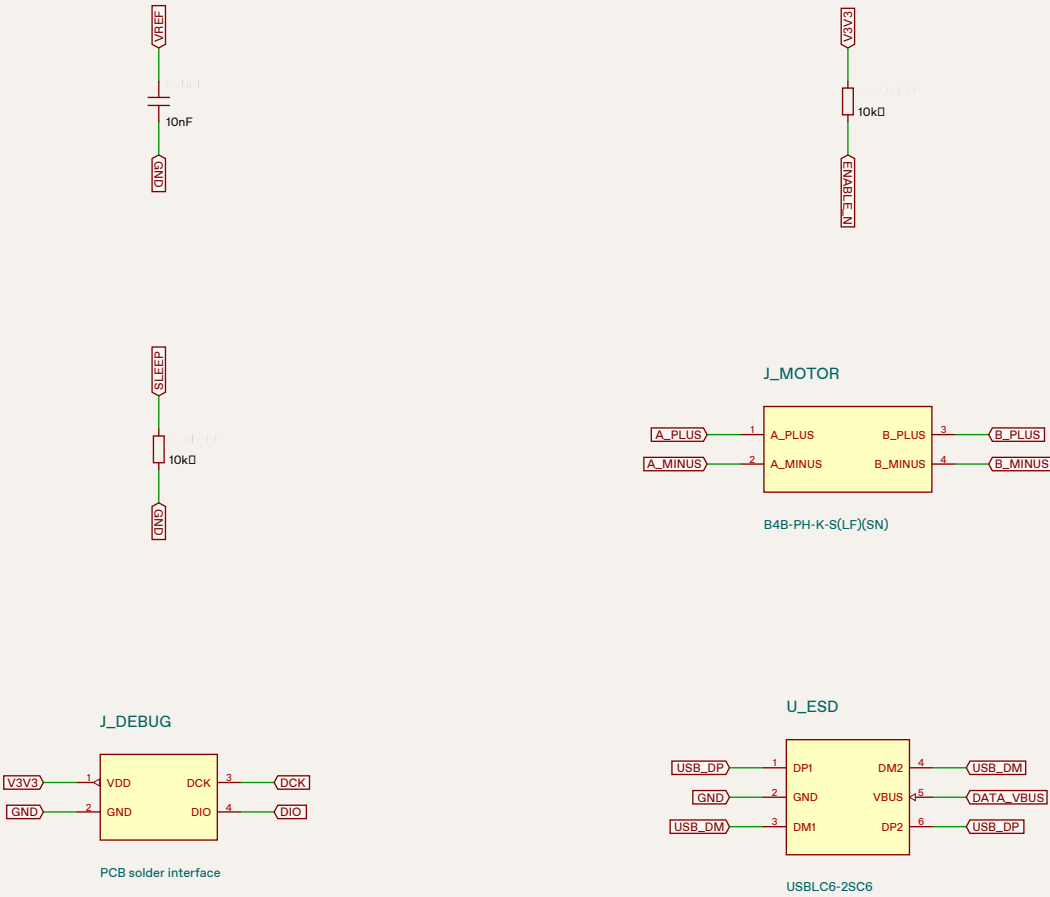
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125

Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F

Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

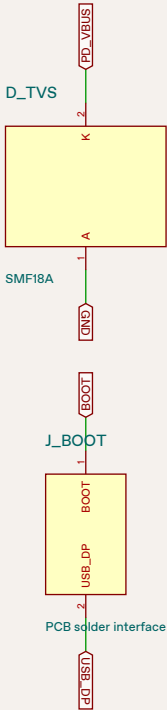


Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.

NEMA14_CH32X035G8U6 - USB-C NEMA14 MOTOR CONTROLLER

SHEET 10/10 - D_TVS, R_USB_BOOT, J_BOOT



COMPONENT PURPOSE

U_MCU - CH32X035G8U6
USB computer control; generates STEP/DIR.
Checks PD voltage before enabling the motor.
Internal clock avoids an external crystal.

U_PD - CH224K
15 V request: CFG1=0, CFG2=CFG3=1.
PG is open-drain, active low. PD-only mode:
chip DP/DM shorted; VBUS sense pin unused.

U_LDO - HT7533-1
Creates 3.3 V from diode-OR USB/PD power.
Low-current LDO retained for prototype only;
measure input spikes, current and temperature.

U_DRV - A4988
Fixed 1/16 steps; 10k enable/sleep defaults.
3.9k/1k REF divider, 0.24 ohm sense resistors:
0.351 A nominal; measure peak phase current.

U_ESD - USBLC6-2SC6
Clamps ESD on USB D+ and D- at the data port.
VBUS pin references the USB protection diodes.

U_VM_ISO - SN74CBTLV1G125
Isolates VM ADC when logic power is off.
10k OE pull-up defaults off; MCU gates ON.
Bus/Z1 when ON; wait 0.5 ms then sample.

Q_DATA - MMBT3904-7-F
Detects host 5 V without direct GPIO feed.
10k base feed and 10k 3.3 V collector pull.
DATA_PRESENT LOW means host attached.

Named nets connect across all pages. NC = intentionally unconnected pin.

PD PORT: motor power, 15 V requested. DATA PORT: USB computer control.
Motor: 14HM11-0404S, bipolar, 0.4 A/phase. Prototype: firmware/bench/rear-fit validation pending.